Appendix J

MITCHELL ACT HATCHERY EIS SOCIOECONOMICS IMPACT METHODS APPENDIX

Appendix – Socioeconomics Impact Methods

This appendix describes the methods and data used to conduct the analysis of socioeconomic effects described in Section 4.3. The analysis of socioeconomic impacts considers predicted harvest-related effects both within the Columbia River Basin and in other regions where Columbia River stocks contribute, and hatchery operations-related effects, including hatchery production costs, associated with affected salmon and steelhead hatcheries in the Columbia River Basin.

1.0 Harvest- Related Effects of Hatchery Production

An excel workbook with linked worksheets, referred to as the Hatchery Impact Model, was developed by TCW Economics to assess harvest-related and hatchery operations-related effects of the Mitchell Act EIS alternatives. Data and values in the worksheets are organized by economic impact regions. The analytical purpose of these regions is to measure the economic impacts (i.e., generation of jobs and personal income) of fishing activity (and, in the case of the Columbia River Basin, hatchery operations as well) that occurs in nearby fisheries. The Columbia River Basin is comprised of four economic impact regions: Lower Columbia River, Mid Columbia River, Upper Columbia River, and Lower Snake River economic impact regions. The Ocean and Puget Sound area includes six economic impact regions: Oregon Coast, California Coast, Washington Coast, Puget Sound/Strait of Juan de Fuca (Puget Sound), British Columbia, and Southeast Alaska (SEAK) economic impact regions.

Each economic impact region in the Columbia River Basin includes a set of counties in which affected fisheries are located. The counties that comprise each economic impact region are as follows:

- **Lower Columbia River economic impact region**: Columbia (OR), Multnomah (OR), Washington (OR), Clackamas (OR), Yamhill (OR), Polk (OR), Marion (OR), Benton (OR), Linn (OR), Lane (OR), Clatsop (OR), Clark (WA), Wahkiakum (WA), Cowlitz (WA), Lewis (WA), Pacific (OR)

- **Mid Columbia River economic impact region**: Hood River (OR), Wasco (OR), Sherman (OR), Gilliam (OR), Morrow (OR), Umatilla (OR), Grant (OR), Wheeler (OR), Crook (OR), Jefferson (OR), Deschutes (OR), Skamania (WA), Klickitat (WA), Bento (WA), Franklin (WA), Walla Walla (WA), Adams (WA)

- **Upper Columbia River economic impact region**: Okanogan (WA), Chelan (WA), Douglas (WA), Kittatas (WA), Yakima (WA)

- **Lower Snake River economic impact region**: Lemhi (ID), Custer (ID), Valley (ID), Adams (ID), Idaho (ID), Clearwater (ID), Lewis (ID), Nez Perce (ID), Latah (ID), Shasone (ID), Wallowa (OR), Union (OR), Asotin (WA), Columbia (WA), Garfield (WA), Whitman (WA)

Commercial (tribal and non-tribal) and recreational fishing activity in affected fisheries (including the mainstem Columbia River and its tributaries) were assigned to the economic region where the fishing activity occurs. The correspondence between fishing areas (both mainstem Columbia River and terminal areas) and economic impact regions in the Columbia River Basin is shown in Figure A-1.

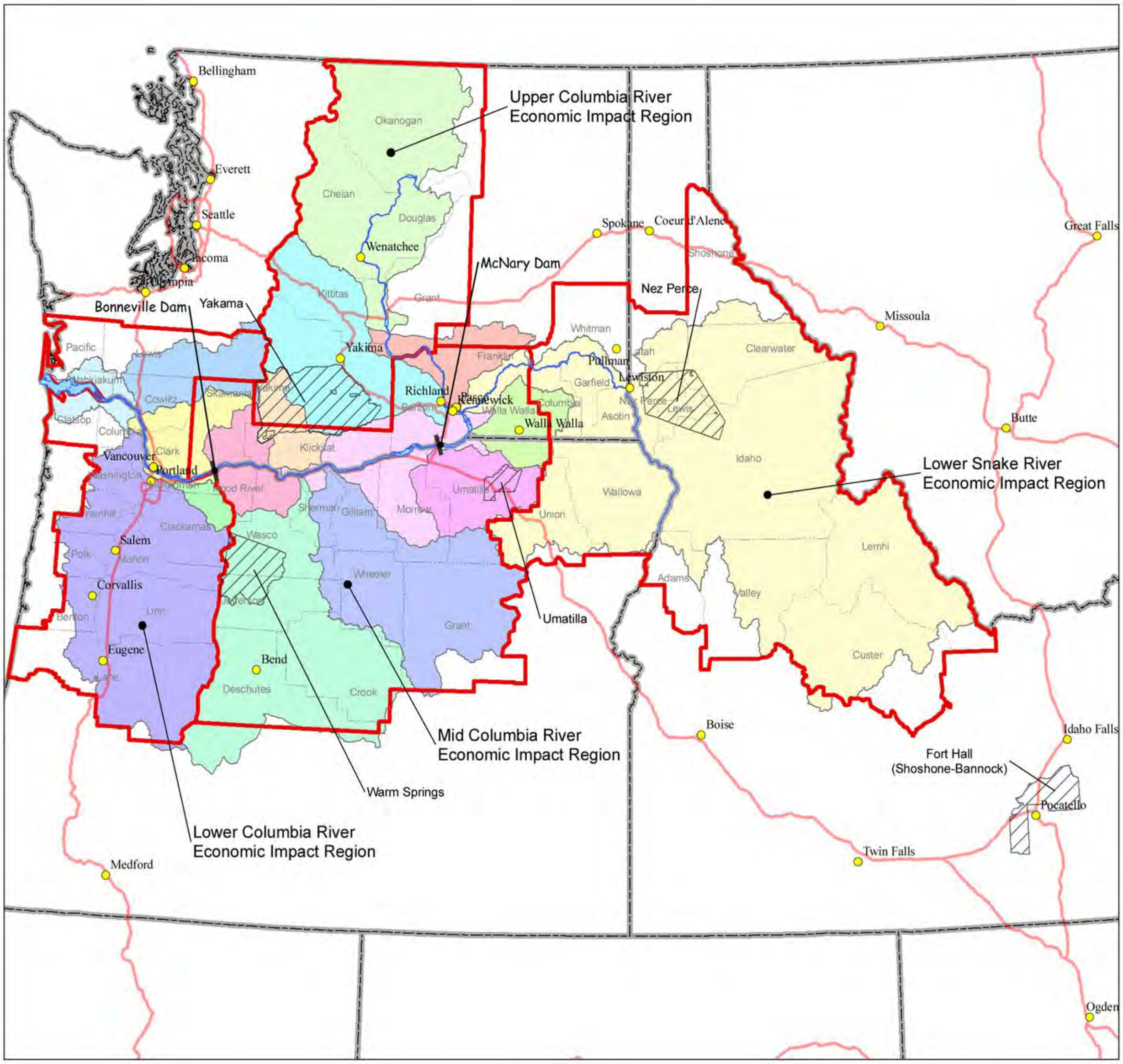

Figure A-1. Terminal Fishing Areas and Economic Impact Regions in the Columbia River Basin

Terminal Fishing Areas

- Columbia Gorge
- Columbia River Estuary
- Cowlitz / Toutle River
- Deschutes River
- Hanford Reach
- John Day River
- Klickitat River
- Lewis / Kalama River
- Lower Mid-Columbia
- Sandy River
- Snake River and Tribs
- Umatilla River
- Upper Columbia
- Walla Walla River
- Willamette River and Tribs
- Yakima River

- Major Cities
- Tribal Reservations
- Interstate Highway

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The six economic impact regions in the Pacific Ocean and Puget Sound Area were defined for purposes of assigning tribal, commercial and recreational fisheries to geographic areas where fisheries affected by Columbia River stocks are located. Because fishing activity associated with the catch of Columbia River stocks occurs almost exclusively in marine waters of these areas, the economic regions are associated with most fishing-related commerce that is generated by fishing activity in these areas. The Oregon, Washington, and California coast economic impact regions are comprised of the coastal counties in those areas, whereas the Puget Sound, British Columbia, and Southeast Alaska regions are less well defined but intended to capture fishing-related economic activity in nearby affected marine waters.

As explained in the Chinook and Coho Salmon Fishery Modeling Approach for Application to the Mitchell Act EIS document (see Appendix K), ocean Chinook salmon fisheries south of Cape Falcon were not modeled as part of the harvest analysis. Chinook fisheries south of Cape Falcon are managed to protect ESA-listed Sacramento River winter and California coastal Chinook and to achieve fall Chinook spawning escapement goals for the Klamath, Sacramento, and Oregon coastal rivers (PFMC 2004, 2005, 2007). Since the abundance of those stocks was fixed in this analysis, and since the migration pattern of Columbia River Chinook is predominantly northward, the potential impact of EIS alternatives on Chinook fisheries south of Cape Falcon would be negligible; therefore, potential harvest-related effects were not quantitatively analyzed in the Socioeconomics and Environmental Justice sections. Section 3.3.5.2 and Table 3-10 in the EIS indicate that Columbia River stocks do not substantially contribute to either the Chinook or coho commercial fisheries south of Cape Falcon.

It should be noted that the lower Columbia River economic impact region includes some counties that also are included in the Oregon and Washington Coast economic regions. This is because salmon fishing activity that occurs in the Lower Columbia River directly contributes fishing-related economic activity to Clatsop County in Oregon and Pacific County in Washington, as does fishing activity along the coastal areas. The lower Columbia River economic region and the Oregon and Washington Coast economic regions, however, are the only regions that have counties common to more than one region (i.e., they geographically overlap).

1.1 Catch Estimates

The Mitchell Act Fishery Modeling Team estimated salmon (coho, Chinook, and sockeye) and steelhead harvest in tribal, non-tribal commercial and recreational fisheries throughout the study area. Catch estimates for each of the EIS alternatives were provided to the Socioeconomics Team. The modeling methods and data used to develop these harvest estimates are described by Lestelle and Morishima (2013). Because the estimates of catch under the alternatives are based on different exploitation rates under the alternatives, fishing activity associated with catch include consideration of changes in both hatchery production and exploitation rates. However, because it is assumed that the underlying dynamics of the fisheries do not appreciably change between the alternatives, the resulting allocation of effort also does not appreciably change as a result of different exploitation rates.

Harvest estimates for fisheries in the Columbia River Basin were based on production and exploitation rates from the late 2000's (refer to the revised Appendix K for details). The predicted number of fish (both wild and hatchery fish) caught in tribal, non-tribal commercial, and recreational fisheries was estimated for areas of the mainstem Columbia River and for different terminal areas within the Columbia River Basin (see Figure A-1). The catch estimates were then assigned to the four economic regions within the Columbia River Basin based on the county (and region) corresponding to the location of the fisheries.

For regions in the Ocean and Puget Sound Area, species-specific catch estimates were provided by the Mitchell Act Fishery Modeling Team for treaty troll, non-treaty troll, and sport fishing fisheries north and south of Cape Falcon. For the marine fisheries off of Oregon, California, and Washington, catch was assigned to port areas based on the average proportion of catch in each port between 2001 and 2005 (see Lestelle and Morishima 2013). For the SEAK, British Columbia, and Puget Sound regions, modeled estimates of total treaty (Puget Sound only), non-tribal commercial, and sport catch (including contributions from the Columbia River and all other river systems) had to be allocated among the different user groups. For alternatives other than Alternative 1, the percentage distribution of catch among each user group over the 2002-2006 period was used for this allocation, supplemented by other information generated by the EIS fishery assessment. The resulting distribution for coho and Chinook catch in these more distant fisheries was as follows:

Coho

- Northwest and southwest Vancouver Island troll: 100% non-tribal commercial
- West coast Vancouver Island: 100% sport
- Puget Sound/Strait of Juan de Fuca: 82% commercial (split 81.6% tribal and 18.4% non-tribal), 18% sport

Chinook

- SEAK: 82% non-tribal commercial, 18% sport
- North and central coasts of British Columbia: 70% non-tribal commercial, 30% sport
- West coast Vancouver Island: 73% non-tribal commercial, 27% sport
- Puget Sound/Strait of Juan de Fuca: 74% commercial (split 88.5% tribal and 11.5% non-tribal), 26% sport

For Alternative 1, which serves as the baseline for analyzing socioeconomic impacts, historical catch (2002-2009 averages) rather than modeled catch estimates were used to characterize catch conditions in the Puget Sound, British Columbia and SEAK regions. Historical averages were used in order to more accurately establish baseline conditions for evaluating the relative effects of changes in harvest conditions compared to the baseline. It should be noted that the tribal harvest in British Columbia and SEAK was not estimated in the harvest analysis for all alternatives.

For the lower Snake River tribal fishery, historical catch data (2008-2011) provided by the Nez Perce Tribe was used to characterize baseline conditions rather than modeled catch estimates. Modeled catch estimates were then used to estimate catch for Alternatives 2 through 6 according to the following steps.

<u>Step 1.</u> Baseline (Alternative 1) catch estimates by species and run were developed based on the average of total harvests (commercial plus C&S) provided by the Nez Perce Tribe for the 2008-2011 period (J. Dixon, pers. comm., NOAA Fisheries West Coast Region, June 16, 2014), as follows:

Spring/Summer Chinook: 9,151
Fall Chinook: 253
Coho: 25

Steelhead: 2,019

Step 2. Total catch (commercial plus C&S) was estimated for the alternatives by developing factors reflecting the percentage change in harvests from Alternative 1 to each alternative, based on modeled total harvest estimates developed by the Mitchell Act Fishery Modeling Team. These percentages, showing the percentage change in total catch relative to Alternative 1, are as follows:

	Alternative 2	Alternative 3	Alternative 4	Alternative 5	Alternative 6
Chinook (all species)	-11.1%	-10.4%	-10.4%	+9.2%	+5.9%
Coho	No modeling information was available for coho. It was assumed not to change.				
Steelhead	+0.1%	+0.2%	+0.3%	+0.2%	+0.1%

These factors were then applied to the baseline (Alternative 1) total catch estimates to produce the following harvest estimates for each alternative.

Species	Alternative 1	Alternative 2	Alternative 3	Alternative 4	Alternative 5	Alternative 6
Spring/Summer Chinook	9,151	8,135	8,199	8,199	9,993	9,695
Fall Chinook	253	225	227	227	276	268
Coho	25	25	25	25	25	25
Steelhead	2,019	2,021	2,023	2,025	2,023	2,021
Total	11,448	10,406	10,474	10,476	12,317	12,009

1.2 Gross and Net Economic Values

1.2.1 Commercial Fisheries

(Note that this section includes analysis of commercial and subsistence harvest because estimates of this harvest are needed to derive estimates of the tribal commercial harvest.)

Estimates of total tribal and non-tribal commercial catch provided by the Mitchell Act Fishery Modeling Team were converted to gross and net economic values using different factors. For estimating gross economic values (ex-vessel values), the number of fish caught was first converted to pounds. The pounds-per-fish factors by species and region used in the conversion are presented in Table A-1. The data sources for these conversion factors include the following:

- Commercial weights (dressed weight per fish) for Washington and Oregon coastal regions: PFMC 2009 SAFE Report, Appendix D, Tables D-2 and D-3 (average weights over the 2002-2009 period)

- Commercial weights (round weight per fish) for Columbia River regions: Calculated based on landings and weight data from fish receiving tickets reported by the Oregon Department of Fish and Wildlife, Columbia River Fishing Landing Reports, 2003-2009, available at http://www.dfw.state.or.us/fish/OSCRP/CRM/Comm_fishery_updates.asp (accessed on December 7, 2011). Calculated weights for each species, including spring, summer, and fall Chinook, were averaged over the 2003-2009 period, weighted by the number of fish landed each year in Oregon. (Note that data were not available for 2002.)

- Commercial weights (round weight per fish) for SEAK and British Columbia regions: TRG 2009, Table B.2.
- Commercial weights (dressed weight per fish) for Puget Sound: Washington Department of Fish and Wildlife (in an excel file provided to the EIS Socioeconomics Team)

Once commercial catch was converted to pounds, the share of catch assumed for ceremonial and subsistence (C&S) purposes was estimated and subtracted from the total tribal harvest in the mid Columbia River upper Columbia River, and lower Snake River regions. Within the Columbia River basin, C&S harvest of salmon occurs both in the basin's mainstem and terminal areas of the economic impact regions. (Note that no quantifiable levels of C&S harvest occur in the lower Columbia River economic impact region.) Although C&S harvest can include coho, steelhead, and summer and fall Chinook, harvest is typically focused on spring Chinook. C&S harvests generally do not vary a great deal from year to year because fish are taken by a tribe to meet the need that a given number of people have for fresh fish; hence, subsistence fish are, in practice, the priority fish taken by a tribe (L. Lestelle, pers. comm., Biostream Environmental, March 28, 2012).

A great deal of uncertainty exists regarding levels of C&S harvests by tribes in the Columbia River Basin. No comprehensive harvest data for past C&S catch in the Columbia River mainstem (Zone 6) are available. In an attempt to collect data on C&S catch, the Mitchell Act Fishery Modeling Team contacted the Columbia River Inter-Tribal Fish Commission (CRITFC) and was told that no estimates of total C&S catch are available (email communication from Stuart Ellis, management biologist for CRITFC, to L. Lestelle, October 25, 2011, as cited in L. Lestelle, pers. comm., Biostream Environmental, October 25, 2011). CRITFC's harvest monitoring system maintains weekly estimates of total catch (commercial plus C&S) of all species but does not keep track of the disposition of fish. As a result, CRITFC was unable to provide an estimate of the size of the tribes' C&S catch relative to their overall catch.

Despite the lack of comprehensive data on C&S harvests in the Columbia River mainstem, the Mitchell Act Fishery Modeling Team estimated average annual C&S harvest by tribes in the mainstem Zone 6 (Middle Columbia River region) fishery (L. Lestelle, pers. comm., Biostream Environmental, March 28, 2012). These estimates are shown in Table 3-26 of the EIS. It should be noted that because of data limitations, actual C&S harvest is likely greater than the estimates in Table 3-27. As a result, the estimates in Table 3-27 should be regarded as minimums. The estimates were developed based on the following data sources:

- *Spring Chinook.* Estimates represent the average of the C&S and platform catch for spring Chinook from 2008-2011, as reported in Table 7 of the most recent Joint Columbia River Management Staff reports for spring fisheries. This time period was selected because it covers the most recent management agreement for U.S. v. Oregon.
- *Summer Chinook.* Estimates represent an average of the C&S catch of summer Chinook over the 2006-2010, as reported in by the Pacific Fisheries Management Council in Table B-20 in its annual review of ocean salmon fisheries. This time period captures the most recently reported catch data.

- *Fall Chinook.* Estimates represent an average of the C&S catch of fall Chinook over the 2003-2007, as reported by the Pacific Fisheries Management Council in Table B-20 in its annual review of ocean salmon fisheries. The reported catch dropped substantially following 2007, likely due to how data was being reported for specific times and areas. As a result, the 2003-2007 period was deemed to be more representative of harvests.
- *Coho.* Estimates represent the average of the C&S catch of coho from 2002-2006, as reported in Joint Columbia River Management Staff reports. (Later reports do not include catch estimates.)

Similar to data for C&S catch in the mainstem Columbia River, comprehensive data for catch in the Columbia River basin's terminal areas are largely lacking across the basin. In an attempt to estimate terminal area catch, several tribes were contacted by the Mitchell Act Fishery Modeling Team to gather information on C&S catch in terminal areas (G. Blair, pers. comm., ICF International, May 9, 2012). Based on information collected from tribes and from other sources, including Joint Columbia River Management Staff reports, the following catch-allocation assumptions for terminal areas were developed and applied to the harvest estimates prepared for Alternative 1 by the Mitchell Act Fishery Modeling Team, resulting in the terminal-area C&S harvest estimates shown in Table 3-26:

- *Columbia Gorge:*

  Spring Chinook: 95% C&S, 5% commercial
  Fall Chinook: 5% C&S, 95% commercial
  Coho: 5% C&S, 95% commercial
  Steelhead: 5% C&S, 95% commercial
- *Klickitat River:*

  Spring Chinook: 95% C&S, 5% commercial
  Fall Chinook: 5% C&S, 95% commercial
  Coho: 5% C&S, 95% commercial
  Steelhead: 5% C&S, 95% commercial
- *Deschutes River:*

  Spring Chinook: 100% C&S, 0% commercial
  Steelhead: 100% C&S, 0% commercial
- *John Day River:*

  Spring Chinook: 100% C&S, 0% commercial
- *Umatilla River:*

  Spring Chinook: 100% C&S, 0% commercial
  Fall Chinook: 100% C&S, 0% commercial
- *Yakima River:*

  Spring Chinook: 100% C&S, 0% commercial
  Summer Chinook: 100% C&S, 0% commercial
  Fall Chinook: 100% C&S, 0% commercial
- *Upper Columbia River (mainstem upstream of Priest Rapids Dam and tributaries):*

  Spring Chinook: 100% C&S, 0% commercial
  Summer Chinook: 100% C&S, 0% commercial
  Steelhead: 100% C&S, 0% commercial

For the lower Snake River mainstem and tributaries, the number of salmon taken for C&S purposes was estimated by assuming that 65.9 percent of all spring/summer Chinook, 0.5

percent of all fall Chinook, and 7.1 percent of all coho caught by tribes is taken for ceremonial and subsistence purposes (L. Lestelle, pers. comm., Biostream Environmental, April 8, 2009). Furthermore, it was assumed that no steelhead would be taken for C&S purposes in the lower Snake River economic impact region.

Considered together, C&S catch from mainstem and terminal areas is estimated, at a minimum, to annually total 19,630 fish in the Middle Columbia River region, with Chinook accounting for 92 percent of the catch (Table 3-26). In the Upper Columbia River Region, C&S catch is estimated to total 2,876 fish. In the lower Snake River economic impact region, C&S catch is estimated to total 6,033 fish, with spring/summer Chinook accounting for virtually all of the catch. These estimates were subtracted from the estimated total tribal harvest in the mid Columbia River, upper Columbia River, and lower Snake River economic impact regions.

Per pound ex-vessel prices for each species and region were then applied to the estimates of the resulting tribal harvest and to the non-tribal commercial harvest to estimate the total regional ex-vessel value of commercial salmon landings in each region. (Note that for all alternatives other than Alternatives 5 and 6, commercial steelhead harvests are estimated to be made only by tribal fishers in the mid Columbia River and lower Snake River economic impact regions. Under Alternatives 5 and 6, a small number of steelhead is also estimated to be harvested by tribal fishers in the upper Columbia River economic impact region.) The value-per-fish factors used to convert estimated landings to total ex-vessel values are shown in Table A-2. The data sources for these value factors include the following:

- Ex-vessel price per pound for Washington and Oregon coastal regions: average prices over the 2002-2009 period calculated based on price data from PFMC 2009 SAFE Report, Tables IV-3 and IV-4.

- Ex-vessel price per pound for Columbia River regions for coho and spring and fall Chinook: calculated based on price and harvest data for Oregon and Washington from PFMC 2009 SAFE Report, Tables IV-8 and IV-9. Prices represent average ex-vessel prices of Columbia River coho and spring and fall Chinook, weighted by pounds of fish landed, over the 2002-2009 period. Fall-season Chinook prices also were weighted by pounds of brights and tules landed in Oregon and Washington each year. (Note that although prices were calculated individually for each species, the same prices were used for tribal fisheries for both the mid and upper Columbia River economic impact regions and for the lower Snake River economic impact region. Typically, prices per pound of fish decline for all species as fish runs move upstream. As a result, prices may be overestimated for the upper Columbia River and lower Snake River tribal fisheries.)

- Ex-vessel price per pound for Columbia River regions for sockeye and summer Chinook: calculated based on price data from Joint Columbia River Management Staff (2010 Joint Staff Report: Stock Status and Fisheries for Spring Chinook, Summer Chinook, Sockeye, Steelhead, and Other Species, and Miscellaneous Regulations), available at http://wdfw.wa.gov/publications/00885/wdfw00885.pdf (accessed on December 12, 2011). Prices represent average of ex-vessel prices for Columbia River sockeye and summer Chinook over the 2008-2009 period. (Data for earlier years was not available.)

- Ex-vessel prices for SEAK, British Columbia, and Puget Sound regions: TRG 2009, Table B.2. Prices were adjusted to 2009 dollars using the gross domestic implicit price index.

Lastly, net economic values (net income) associated with the commercial harvest were estimated. Per-fish factors (Table A-3) derived from TRG 2009, Table B.2., were used to estimate net economic values (Values were adjusted to 2009 dollars using the gross domestic implicit price index). It should be noted that the average net value factors used in the analysis may not fully capture the effects of increasing incremental costs as abundance declines. Accurately capturing this potential effect on net economic values would require a substantially more technically complicated approach to valuation, one that is considered beyond the rigor needed for the comparative analysis conducted for the EIS. As a result, the estimates of changes in the net economic value of the commercial harvest (EIS Table 4-102) may be lower than actual changes, particularly for alternatives with relatively large changes, such as Alternative 2.

1.2.2 Recreational Fisheries

Table A-4 shows the angler-trip conversion factors used to convert catch to angler trips for each species and region. The data sources for these conversion factors include the following:

- Sport catch per trip for SEAK, British Columbia, Washington, Oregon, and California coastal regions: compiled from 2002-2009 salmon landing and effort values from the PFMC 2011 SAFE Report, Tables A-4, A-5, A-9, A-10, A-17, and A-18. (Note that Washington coastal values were used for estimates of sport catch per trip for SEAK and British Columbia.)

- Sport catch per trip for Columbia River region: compiled from 2002-2009 angler trips and catch data from Catch Record Card data provided by WDFW. (Note that sport-catch-per-trip factors were developed for individual species but that the same factors were used for species across all four Columbia River Basin economic impact regions. As a result, while trip estimates for the entire basin may be reasonably reliable, sport trips may be overestimated in some regions and underestimated in others.)

Once catch was converted to sport angler trips, per trip expenditure factors for each species and region were applied to the estimated number of sport trips to estimate the total trip-related expenditures in each region. The per trip expenditure factors, which are shown in Table A-5 in 2009 dollars, were developed based on the following data sources.

- Columbia River regions: derived from data available from the Oregon Angler Survey and Economic Study (The Research Group 1991).

- Puget Sound region: TRG 2009, Table B.2. Prices were adjusted to 2009 dollars using the gross domestic implicit price index.

- SEAK and British Columbia: estimated based on data from the Economic Contribution and Impacts of Salmonid Resources in Southeast Alaska report (TCW Economics 2011).

- Washington, Oregon, and California: estimated based on data from 2006 NOAA National Marine Recreational Fishing Expenditure Survey.

Net economic values (willingness to pay for fishing over and above expenditures) associated with the recreational fishery were estimated using per angler day values derived from a review of past studies of anglers' net willingness to pay for salmon fishing in the Pacific Region and Alaska (Boyle et. al 1998). These factors, which were adjusted to 2009 dollars using the gross domestic implicit price index, are $41.02 per angler day for fishing in Alaska and $61.07 for salmon fishing in all other areas. For the analysis, it was assumed that an angler trip is equivalent to an angler day.

It should be acknowledged that the relationship between angler effort and economic values (both gross and net economic values) is not linear. The precise form of this relationship, however, varies depending on many relevant factors, including baseline catch per unit of effort conditions, the magnitude of change in fish populations, and the mode of fishing/locational factors. The effect that changes in recreational catch (as influenced by hatchery production, among other factors) have on angler effort and economic values varies across a relatively wide range of studies of angler activity. Although changes in fish populations have variable effects on angler effort and associated economic values depending on the magnitude of fish population changes, accurately estimating these effects across the entire range of population changes (as assumed for the different alternatives) requires a more robust analysis than could be developed for the assessment in the EIS. Consequently, point estimates (i.e., values that do not vary across the range of population changes) of both net WTP and trip expenditures had to be used with estimates of fishing effort derived from the predicted changes in the number of fish caught provided by the Mitchell Act Fish Modeling Team.

1.3 Harvest-Related Regional and Local Economic Impacts

Harvest-related regional economic impacts are generated by three fishery components: 1) economic activity from tribal commercial harvests, 2) economic activity from non-tribal commercial harvests, and 3) economic activity generated by sport fishing. Estimates of regional economic impacts from these activities are expressed in terms of personal income and jobs generated in each of the 10 regions in the Columbia River Basin and the Pacific Ocean and Puget Sound Area.

1.3.1 Personal Income

To estimate total (direct, indirect, and induced) personal income generated by estimated commercial and recreational catch under each alternative, personal income impact factors for each species and region were applied to the converted catch (i.e., pounds of commercial landings and sport trips). Table A-6 shows the regional personal income impact factors (in 2009 dollars) used to convert catch (in pounds) and angler trips for each user group, species, and region to personal income impacts. The sources for the regional income impact factors include the following:

- Regional income impact factors (per pound) for commercial catch in the Puget Sound region and the Washington and Oregon coastal regions: Fishery Economic Assessment Model (FEAM) impact factors for 2007 provided by the PFMC in file "Tables CH IV Econ Sup." (These factors were those used to produce the regional economic impacts presented in the PFMC 2008 SAFE Report.)

Regional income impact factors (per pound) for commercial catch in SEAK, British Columbia, and Columbia River Basin regions: TRG 2009, Table B.2.

- Sport income impact factors (per angler trip) for Puget Sound region and the Washington, Oregon, and California coastal regions: FEAM charter and private boat trip impact factors for 2007 provided by the PFMC in file "Tables CH IV Econ Sup" (factors for charter and private boats were weighted based on boat-type trip distributions over the 2004-08 period for each region, as reported in the PFMC 2008 SAFE Report).

Sport income impact factors (per angler trip) for SEAK, British Columbia, and Columbia River Basin regions: TRG 2009, Table B.2. It should be noted that regional income is measured as personal income accruing to households. It measures the contribution to personal income under current (or changed) conditions. Because dynamic changes in the economy over time are not considered in this analysis, results of the assessment are not considered valid for measuring effects on the economy over the long term from changes in fish abundance or policy.

1.3.2 Jobs

Jobs (full- and part-time; direct, indirect, and induced) generated by the commercial and recreational catch in each region under each alternative were estimated by applying an earnings-per-job factor to the estimated total personal income generated by catch in each region described above. The earnings-per-job factors for each region were calculated by dividing total earnings in each region in 2007 by total jobs, as reported by the Bureau of Economic Analysis (BEA) (BEA Table CA05N: Personal Income by Major Source and Earnings by NAICS Industry; BEA Table CA25N: Total Full- and Part-Time Employment by NAICS Industry). For California, Oregon and Washington Coast economic regions, factors were developed for individual counties that comprise the regions. The resulting earnings-per-job factors are presented in Table A-7. (Note that the earnings-per-job factor for the Puget Sound region also was used for SEAK and British Columbia.) The personal income totals for each region were then divided by the earnings-per-jobs factors to estimate jobs for each region and alternative.

2.0 Hatchery-Operations Related Effects

Although the analysis of socioeconomic effects focused on harvest-related effects from expected changes in Columbia River Basin hatchery production, operational effects, including effects on production costs, hatchery jobs and personal income generated by production changes, also were evaluated. This section describes the methods and data used to conduct these analyses.

2.1 Cost Analysis

Smolt production at salmon and steelhead hatcheries in the Columbia River Basin varies under the EIS alternatives; consequently, hatchery production costs also vary by alternative. The assessment of hatchery operations costs considers baseline costs associated with Alternative 1 (No Action) and changes in these baseline costs associated with changes in smolt production (and release), implementation of best management practices (BMPs), and construction and operation of new weirs. Baseline costs include smolt production costs, indirect or overhead (i.e.,"headquarters") costs, and amortized capital costs associated with hatchery facility improvements. Changes in hatchery operations costs under the project alternatives considers only variable costs (i.e., those costs that change in response to smolt production changes), and additional (incremental) costs for implementing BMPs and constructing and operating new weirs.

Smolt production costs were estimated separately for facilities funded by the Mitchell Act and for other hatchery facilities in the Columbia River Basin that produce salmon and steelhead. Budget data compiled by NOAA Fisheries (personal communication with Allyson Purcell, National Marine Fisheries Service, June 22, 2009) for hatcheries funded by the Mitchell Act were used to estimate smolt production costs for both Mitchell Act and non-Mitchell Act hatchery facilities. Facility-specific and average smolt production costs from Mitchell Act hatcheries are presented in Table A-8. Average smolt production costs by entity and species in Table A-8 were primarily used to estimate costs for non-Mitchell Act hatchery facilities in the Columbia River Basin that would be affected.

Smolt production costs were estimated by multiplying the estimated number of smolts produced at each hatchery by a unit cost factor (cost per smolt) developed for each hatchery and species, for both Mitchell Act and non-Mitchell Act hatcheries. The unit cost factors are composed of variable operating costs and fixed costs. Fixed costs include the unit costs attributable to administration of the hatchery programs at the agency headquarters level and short-term hatchery capital costs (facility replacement). For Alternative 1 (the baseline condition), hatchery production costs were estimated based on variable operating and fixed costs associated with production levels for each hatchery and species under Alternative 1. For Alternatives 2 through 5, smolt production costs were estimated by applying the variable operating costs to the smolt production levels specific to each hatchery under each alternative, then adding in the fixed costs associated with Alternative 1. (This reasonably assumes that fixed costs associated with baseline conditions would not change in response to changes in smolt production levels under each alternative.) For a small number of hatcheries, no smolts are predicted to be produced under baseline conditions (Alternative 1), suggesting that fixed costs for these hatcheries would be zero. For these hatcheries, the fixed-cost component of the unit cost factor was assumed to be the same as the fixed cost for the alternative with the highest smolt production levels across the five action alternatives. (The excel spreadsheet used to calculate costs by region is available upon request.)

Costs for implementing BMPs and constructing and operating new weirs were developed by the Mitchell Act EIS team. The purpose of the cost estimating was to develop capital costs at a pre-conceptual level for BMPs identified in the region. Where available, HGMPs were reviewed and information was used to obtain production levels and estimated water needs to help identify the general scope of specific BMPs. A pre-conceptual cost estimation process was conducted to identify probable ranges of costs for capital needs identified for the BMPs. This was done by applying probable costs from similar projects. The analysis was completed without review of facility drawings or review of specific facilities. No sites were visited to specifically verify costs applied to each BMP.

Key considerations of the estimates of BMP costs include:

- All costs for adherence to BMPs are included. In some cases, BMPs do not have any capital costs associated with them. Fish passage, however, might be an exception.

- The cost to provide fish passage at hatcheries that currently block migration was not estimated. Fish passage costs were not developed because it was determined that they would vary greatly depending on the specific site constraints, total flow requirements, facility size and location and related unforeseeable implementation issues.

- The cost of any additional staffing associated with meeting BMPs (with the exception of staff to provide security) was not included.

For the analysis, it was assumed that BMP and weir costs would be as follows:

- A water supply alarm would cost an estimated $10,000.
- A back-up power generator is estimated to cost between $30,000 and $50,000.
- An updated water intake screen is estimated to cost between $200,000 to $500,000, depending on the specific site constraints, total flow requirements, facility size, location and related unforeseeable implementation issues.
- Around-the-clock staffing for security reasons would cost $100,000 per year.
- Installing a water treatment system to ensure pathogen-free water is estimated to cost between $100,000 to $1,000,000, depending on the specific site constraints, total flow requirements, facility size and location, and related unforeseeable implementation issues.
- Fixing water intake structures is estimated to cost between $50,000 and $1,000,000 depending on facility size and location, specific site constraints, total flow requirements, location and related unforeseeable implementation issues.
- Construction of small weir systems is estimated to cost between $250,000 and $300,000 per weir; construction of medium weir systems is estimated to cost between $300,000 and $350,000 per weir; and construction of large weir systems is estimated to cost $500,000 per weir.
- Annual weir operating costs are estimated to average $100,000.

It should be noted that actual costs for BMPs would depend on specific site constraints, total flow requirements, site location, and related unforeseeable implementation issues.

The estimates of BMP costs provide a very general benchmark for future planning, and broad assumptions for overall costs in a region. To develop actual implementation costs for a specific BMP, preliminary and final design processes would be needed, and associated annual operations and maintenance costs should be considered. Other cost considerations, such as sequencing, economy of scale, and inflation and escalation, also would need to be considered. The costs estimated through this exercise should not be utilized for obtaining funding for implementing a specific BMP. This would require the review of specific facility drawings and documents, and investigating the specific items by an engineer.

The full costs of implementing hatchery reform were not included in the analysis. The cost estimates only allow for a relative comparison of costs across alternatives. As decisions are made on priorities for implementation of hatchery reform, the probable ranges of costs provided would be refined through standard planning and design processes. Estimates of the full cost of hatchery reform are not currently available.

Lastly, one-time costs for BMPs and new weirs by alternative were annualized using a 3.8 percent annual amortization rate, which was the current discount rate recommended by the Office and Management Budget, over a 30-year amortization period.

2.2 Effects on Regional Economic Activity

Hatcheries support jobs in the Columbia River Basin economy by directly employing workers, and from economic activity generated by procuring goods and services needed for hatchery operations from regional businesses. Expenditures on hatchery labor and the procurement of goods and services produce indirect and induced effects on employment and personal income in regional economies.

The analysis of hatchery operations-related effects was based on the estimates of annual hatchery costs for each alternative. Estimates of hatchery production costs (including annualized costs for implementing BMPs and constructing and operating new weirs) were as follows:

- Alternative 1: $83.23 million
- Alternative 2: $59.09 million
- Alternative 3: $83.16 million
- Alternative 4: $84.74 million
- Alternative 5: $88.86 million
- Alternative 6: $95.07 million

To assess hatchery operations-related effects at the regional level, hatchery facility operation costs within the four Columbia River Basin economic regions were developed. The operating costs, including smolt production costs, BMP costs and new weir construction and operating costs, were assigned to the appropriate regions based on the location of hatchery facilities within each region. (As stated above for the calculation of smolt production costs, the excel spreadsheet used to calculate costs by region is available upon request.)

The number of jobs directly supported by hatchery operations under each alternative was estimated by applying a factor of 8 jobs per million dollars of costs to the estimates of regional hatchery costs. This factor was derived based on a review of budget/jobs relationships from budget information on salmon and steelhead hatcheries operated by WDFW, ODFW, and the Yakama Nation. The total number of jobs (direct, indirect, and induced) generated in each region as a result of direct hatchery employment was estimated using a multiplier of 1.5, which was based on an employment multiplier for Washington State generated by the IMPLAN input-output model (Minnesota IMPLAN Group 2008) for the industrial sector that includes fish hatcheries (Animal Production except for Cattle, Poultry, and Eggs).

Employment generated by hatchery-related procurement expenditures was estimated by assuming that 40 percent of hatchery production costs under each alternative are attributable to procurement expenditures, and that half of procurement expenditures are made locally (i.e., within regional economies). These assumptions were developed based on the professional judgment of TCW Economics following a review of available hatchery budget information. Jobs directly generated by procurement expenditures in each region were estimated using a factor of 10 jobs per million dollars of procurement expenditures. This factor was derived based on employment coefficients (jobs per million dollars of output) produced by the IMPLAN model for Washington State for wholesale and retail trade sectors. The total number of jobs (direct, indirect, and induced) generated in each region by procurement spending was estimated using a multiplier of 1.7, which was derived based on IMPLAN employment multipliers for retail trade sectors.

Personal income attributable to direct, indirect and induced jobs was estimated based on an estimated factor of $50,000 per job. This personal income factor was developed based on two sources. For direct hatchery jobs, a review of salary and benefits data for hatchery jobs provided by hatchery budgets suggested that average employee compensation, including benefits, per direct hatchery job was about $50,000. For procurement-related jobs and indirect and induced jobs generated by both direct hatchery employment and procurement spending, the IMPLAN model's 2007 database for Washington State suggested average personal income, including employee compensation and proprietor income, of about $50,000 per job across all jobs in the state. This factor was applied to estimated jobs to produce estimates of total personal income in each region for each alternative.

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Table A-1. Average pounds per fish (commercial harvest)

	Tribal				Non-Tribal Commercial	
REGION	Coho	Chinook	Steelhead	Sockeye	Coho	Chinook
COLUMBIA RIVER BASIN						
Lower Snake River	10.3		10.6	na	na	
Spring Chinook		14.2				na
Summer Chinook		17.1				na
Fall Chinook		18.3				na
Upper Columbia River	10.3		10.6	na	na	
Spring Chinook		14.2				na
Summer Chinook		17.1				na
Fall Chinook		18.3				na
Mid Columbia River	10.3		10.6	3.5	na	
Spring Chinook		14.2				na
Summer Chinook		17.1				na
Fall Chinook		18.3				na
Lower Columbia River	na		na	na	9.4	
Spring Chinook		na				14.1
Summer Chinook		na				18.8
Fall Chinook		na				19.1
OCEAN AND PUGET SOUND REGION						
Oregon Coast						
Astoria (Clatsop)	na	na	na	na	7.2	11.9
Tillamook (Tillamook)	na	na	na	na	na	na
Newport (Lincoln)	na	na	na	na	na	na
Coos Bay (Coos)	na	na	na	na	na	na
Brookings (Curry	na	na	na	na	na	na
California Coast						
Crescent City (Del Norte)	na	na	na	na	na	na
Eureka (Humboldt)	na	na	na	na	na	na
Fort Bragg (Mendocino)	na	na	na	na	na	na
San Francisco (San Francisco)	na	na	na	na	na	na
Monterey (Monterey)	na	na	na	na	na	na
Washington Coast						
Neah Bay (Clallam)	6.3	10.1	na	na	6.7	13.1
LaPush (Jefferson)	6.3	10.1	na	na	6.7	13.1
Westport (Grays Harbor)	6.3	10.1	na	na	6.7	13.1
Ilwaco (Pacific)	6.3	10.1	na	na	6.7	13.1
Puget Sound/SJDF	6.4	10.8	na	na	6.4	10.8
British Columbia	na	na	na	na	5.5	17.7
Southeast Alaska	na	na	na	na	6.5	17.5

Notes:
na = not applicable

Sources:
Commercial weights (round weight) for SEAK and BC: TRG 2009, Table B.2.
Commercial weights (round weight) for Columbia River Basin regions: average weights over the 2003-2009 period, weighted by number of fish landed each year in Oregon (or averaged over years for which data were available). Note that data for 2002 were not available. Calculated based on landings and weight data from fish receiving tickets reported by Oregon Department of Fish and Wildlife. 2011. Columbia River Fishing Landing Reports, 2003-2009. Available at http://www.dfw.state.or.us/fish/OSCRP/CRM/Comm_fishery_updates.asp. Accessed on December 7, 2011.
Commercial weights (dressed weight) for Puget Sound/SJDF: Washington Department of Fish and Wildlife (Excel File="PS Salmon_average weight per fish.xlsx").
Commercial weights (dressed weight) for Washington and Oregon regions: average weights over the 2002-2009 period (Pacific Fishery Management Council 2010 (Review of 2009 Ocean Salmon Fisheries, Tables D-2 and D-3).

Table A-2. Ex-vessel price per pound (2009 dollars)

	Tribal				Non-Tribal Commercial	
REGION	Coho	Chinook	Steelhead	Sockeye	Coho	Chinook
COLUMBIA RIVER BASIN						
Lower Snake River	$0.70		$0.67	na	na	
Spring Chinook		$2.67				na
Summer Chinook		$2.77				na
Fall Chinook		$0.85				na
Upper Columbia River	$0.70		$0.67	na	na	
Spring Chinook		$2.67				na
Summer Chinook		$2.77				na
Fall Chinook		$0.85				na
Mid Columbia River	$0.70		$0.67	$2.58	na	
Spring Chinook		$2.67				na
Summer Chinook		$2.77				na
Fall Chinook		$0.85				na
Lower Columbia River	na		na	na	$0.91	
Spring Chinook		na				$4.22
Summer Chinook		na				$2.77
Fall Chinook		na				$1.34
OCEAN AND PUGET SOUND REGION						
Oregon Coast						
Astoria (Clatsop)	na	na	na	na	$1.91	$4.46
Tillamook (Tillamook)	na	na	na	na	na	na
Newport (Lincoln)	na	na	na	na	na	na
Coos Bay (Coos)	na	na	na	na	na	na
Brookings (Curry	na	na	na	na	na	na
California Coast						
Crescent City (Del Norte)	na	na	na	na	na	na
Eureka (Humboldt)	na	na	na	na	na	na
Fort Bragg (Mendocino)	na	na	na	na	na	na
San Francisco (San Francisco)	na	na	na	na	na	na
Monterey (Monterey)	na	na	na	na	na	na
Washington Coast						
Neah Bay (Clallam)	$1.69	$4.18	na	na	$1.97	$3.83
LaPush (Jefferson)	$1.69	$4.18	na	na	$1.97	$3.83
Westport (Grays Harbor)	$1.69	$4.18	na	na	$1.97	$3.83
Ilwaco (Pacific)	$1.69	$4.18	na	na	$1.97	$3.83
Puget Sound/SJDF	$1.67	$1.98	na	na	$1.67	$1.98
British Columbia	na	na	na	na	$1.39	$3.33
Southeast Alaska	na	na	na	na	$0.86	$2.77

Notes:
na = not applicable

Sources:
Ex-vessel prices for SEAK, BC, and Puget Sound/SJDF: Appendix I of the Mitchell Act Public Draft EIS, Table A.3. Prices adjusted to 2009 dollars using gross domestic implicit price index.

Ex-vessel prices for Columbia River Basin coho and spring and fall Chinook: Represents average ex-vessel prices of Columbia River fish, weighted by pounds of fish landed,over the 2002-2009 period. Fall-season Chinook prices also were weighted by pounds of brights and tules landed in Oregon and Washington in each year. Prices calculated based on price and harvest data from Pacific Fishery Management Council 2010 (Review of 2009 Ocean Salmon Fisheries, Tables IV-8 and IV-9).

Ex-vessel prices for Columbia River Basin sockeye and summer Chinook: Represents average ex-vessel prices of Columbia River fish over the 2008-2009 period. Prices calculated based on price data from Joint Columbia River Management Staff 2010 (2010 Joint Staff Report: Stock Status and Fisheries for Spring Chinook, Summer Chinook, Sockeye, Steelhead, and Other Species, and Miscellaneous Regulations. Available at http://wdfw.wa.gov/publications/00885/wdfw00885.pdf. Accessed on December 12, 2011.)

Ex-vessel price for Columbia River Basin Steelhead: TRG 2009, Table B.2. Prices adjusted to 2009 dollars using gross domestic implicit price index.

Ex-vessel prices for Washington and Oregon coatal regions: Represents average price per dressed pound of commercial non-Indian landings over the 2003-2009 period. Prices calculated based on price data from Pacific Fishery Management Council 2010 (Review of 2009 Ocean Salmon Fisheries, Tables IV-3 and IV-4).

Table A-3. Net economic value (net income) factors for commercial fishing (value per fish)

REGION	Coho	Chinook	Steelhead	Sockeye
COLUMBIA RIVER BASIN				
Lower Snake River	$6.62	$22.55	$7.34	na
Upper Columbia River	$6.62	$22.55	$7.34	$6.62
Mid Columbia River	$6.62	$22.55	$7.34	$6.62
Lower Columbia River	$6.62	$22.55	$7.34	$6.62
OCEAN AND PUGET SOUND REGION				
Oregon Coast	$7.44	$33.33	na	na
California Coast	na	na	na	na
Washington Coast	$3.98	$23.16	na	na
Puget Sound/SJDF	$6.67	$18.18	na	na
British Columbia	$2.78	$16.99	na	na
Southeast Alaska	$2.78	$16.99	na	na

Notes:
1. Columbia River Basin Chinook and coho are weighted averages, with weights derived from the relative share of fall and spring/summer chinook harvest and also the relative share of harvest in tribal and non-tribal commercial fisheries.
2. For sockeye, the coho NEV factor was used.
3. na = not applicable

Source:
TRG 2009, Table B.2. (NEV per fish). Prices adjusted to 2009 dollars using the gross domestic implicit price index.

Table A-4. Average catch per recreational fishing trip, by species and region

REGION	Coho	Chinook	Steelhead
COLUMBIA RIVER BASIN			
Lower Snake River	0.24		0.19
Spring Chinook		0.19	
Summer Chinook		0.19	
Fall Chinook		0.23	
Upper Columbia River	0.24		0.19
Spring Chinook		0.19	
Summer Chinook		0.19	
Fall Chinook		0.23	
Mid Columbia River	0.24		0.19
Spring Chinook		0.19	
Summer Chinook		0.19	
Fall Chinook		0.23	
Lower Columbia River	0.24		0.19
Spring Chinook		0.19	
Summer Chinook		0.19	
Fall Chinook		0.23	
OCEAN AND PUGET SOUND REGION			
Oregon Coast			
Astoria (Clatsop)	0.82	0.82	na
Tillamook (Tillamook)	0.82	na	na
Newport (Lincoln)	0.82	na	na
Coos Bay (Coos)	0.82	na	na
Brookings (Curry	0.82	na	na
California Coast			
Crescent City (Del Norte)	0.81	na	na
Eureka (Humboldt)	0.81	na	na
Fort Bragg (Mendocino)	0.81	na	na
San Francisco (San Francisco)	0.81	na	na
Monterey (Monterey)	0.81	na	na
Washington Coast			
Neah Bay (Clallam)	1.22	1.22	na
LaPush (Jefferson)	1.22	1.22	na
Westport (Grays Harbor)	1.22	1.22	na
Ilwaco (Pacific)	1.22	1.22	na
Puget Sound/SJDF	1.22	1.22	na
British Columbia	1.22	1.22	na
Southeast Alaska	1.22	1.22	na

Notes:
na = not applicable

Sources:
Sport catch per trip for Washington, Oregon, and California coastal regions: compiled from 2002-2009 salmon landing and effort values from the 2011 SAFE Report, Tables A-4, A-5, A-9, A-10, A-17, and A-18. Washington coastal values also used for estimating sport catch per trip for British Columbia and Southeast Alaska.
Sport catch per trip for Columbia River: compiled from 2002-09 angler trips and catch data from Sport Catch Record data (Table 2) provided by WDFW (Dixon pers. comm.).

Table A-5. Expenditures per sport trip (2009 dollars)

REGION	Coho	Chinook	Steelhead
COLUMBIA RIVER BASIN			
Lower Snake River	$83.30	$83.30	$83.30
Upper Columbia River	$83.30	$83.30	$83.30
Mid Columbia River	$83.30	$83.30	$83.30
Lower Columbia River	$83.30	$83.30	$83.30
OCEAN AND PUGET SOUND REGION			
Oregon Coast			
Astoria (Clatsop)	$119.71	$119.71	na
Tillamook (Tillamook)	$119.71	na	na
Newport (Lincoln)	$119.71	na	na
Coos Bay (Coos)	$119.71	na	na
Brookings (Curry	$119.71	na	na
California Coast			
Crescent City (Del Norte)	$155.91	na	na
Eureka (Humboldt)	$155.91	na	na
Fort Bragg (Mendocino)	$155.91	na	na
San Francisco (San Francisco)	$155.91	na	na
Monterey (Monterey)	$155.91	na	na
Washington Coast			
Neah Bay (Clallam)	$147.52	$147.52	na
LaPush (Jefferson)	$147.52	$147.52	na
Westport (Grays Harbor)	$147.52	$147.52	na
Ilwaco (Pacific)	$147.52	$147.52	na
Puget Sound/SJDF	$74.47	$73.47	na
British Columbia	$191.79	$191.79	na
Southeast Alaska	$191.79	$191.79	na

Notes:
na = not applicable
Expenditures have been adjusted to 2009 dollars using the gross domestic implicit price index.

Sources:
Expenditures for Columbia River Basin: Oregon Angler Survey and Economic Study, The Research Group 1991.
Expenditures for Puget Sound/SJDF: TRG 2009, Table B.2 (NEV per fish).
Expenditures for SEAK and BC: estimated based on data from TCW Economic's (2010) Southeast Alaska report.
Expenditures for Washington, Oregon, and California: estimated based on data from 2006 NOAA National Marine Recreational Fishing Expenditure Survey.

Table A-6. Personal income factors, per pound of commercially landed salmon and per sport trip (2009 dollars)

	Tribal				Non-Tribal Commercial		Sport		
REGION	Coho	Chinook	Steelhead	Sockeye	Coho	Chinook	Coho	Chinook	Steelhead
COLUMBIA RIVER BASIN									
Lower Snake River	$5.02	$4.31	$4.04	na	na	na	$60.47	$60.47	$60.47
Upper Columbia River	$5.02	$4.31	$4.04	na	na	na	$60.47	$60.47	$60.47
Mid Columbia River	$5.02	$4.31	$4.04	$5.02	na	na	$60.47	$60.47	$60.47
Lower Columbia River	na	na	na	na	$2.59	$3.21	$60.47	$60.47	$60.47
OCEAN AND PUGET SOUND REGION									
Oregon Coast									
Astoria (Clatsop)	na	na	na	na	$2.80	$7.93	$58.29	$58.29	na
Tillamook (Tillamook)	na	na	na	na	na	na	$41.59	na	na
Newport (Lincoln)	na	na	na	na	na	na	$62.66	na	na
Coos Bay (Coos)	na	na	na	na	na	na	$45.65	na	na
Brookings (Curry	na	na	na	na	na	na	$38.24	na	na
Regionwide Total	na	na	na	na	na	na	$60.59	na	na
California Coast									
Crescent City (Del Norte)	na	na	na	na	na	na	$40.06	na	na
Eureka (Humboldt)	na	na	na	na	na	na	$45.28	na	na
Fort Bragg (Mendocino)	na	na	na	na	na	na	$67.05	na	na
San Francisco (San Francisco)	na	na	na	na	na	na	$94.95	na	na
Monterey (Monterey)	na	na	na	na	na	na	$68.14	na	na
Regionwide Total	na	na	na	na	na	na	$83.54	na	na
Washington Coast									
Neah Bay (Clallam)	$2.10	$5.66	na	na	$2.10	$5.66	$39.77	$39.77	na
LaPush (Jefferson)	$2.43	$7.06	na	na	$2.43	$7.06	$48.78	$48.78	na
Westport (Grays Harbor)	$2.10	$8.60	na	na	$2.10	$8.60	$124.65	$124.65	na
Ilwaco (Pacific)	$2.80	$8.16	na	na	$2.80	$8.16	$82.46	$82.46	na
Regionwide Total	$2.54	$7.15	na	na	$2.54	$7.15	$87.59	$87.59	na
Puget Sound/SJDF	$2.44	$7.94	na	na	$2.44	$7.94	$89.72	$89.72	na
British Columbia	na	na	na	na	$1.90	$3.69	$306.94	$297.13	na
Southeast Alaska	na	na	na	na	$2.31	$2.84	$306.94	$297.13	na

Notes:
na = not applicable
All income factors have been adjusted to 2009 dollars using the gross domestic implicit price index.

Sources:
Income factors for SEAK, BC, Columbia River Basin regions: TRG 2009, Table B.2.
Income factors for Puget Sound/SJDF and Washington and Oregon coastal areas: PFMC in file "Tables CH IV Econ Sup". (Sport factors for charter and private boats were weighted based on trip boat-type trip distributions over the 2004-08 period for each port area.)

Table A-7 Earnings per Job (2009 dollars)

REGION	
COLUMBIA RIVER BASIN	
Lower Snake River	$26,220
Upper Columbia River	$30,159
Mid Columbia River	$33,471
Lower Columbia River	$39,461
OCEAN AND PUGET SOUND REGION	
Oregon Coast	
Astoria (Clatsop)	$34,931
Tillamook (Tillamook)	$32,249
Newport (Lincoln)	$31,933
Coos Bay (Coos)	$33,481
Brookings (Curry	$28,034
Regionwide Total	$32,693
California Coast	
Crescent City (Del Norte)	$38,184
Eureka (Humboldt)	$35,420
Fort Bragg (Mendocino)	$33,952
San Francisco (San Francisco)	$85,581
Monterey (Monterey)	$52,330
Regionwide Total	$72,287
Washington Coast	
Neah Bay (Clallam)	$34,109
LaPush (Jefferson)	$30,296
Westport (Grays Harbor)	$37,262
Ilwaco (Pacific)	$30,061
Regionwide Total	$34,193
Puget Sound/SJDF	$57,768
British Columbia	$57,768
Southeast Alaska	$57,768

Notes:

For Southeast Alaska and British Columbia, earnings-per-job for Puget Sound are used.

All income factors have been adjusted to 2009 dollars using the gross domestic implicit price index.

Sources:

Bureau of Economic Analysis. April 2009. Table CA05N Personal Income by Major Source and Earnings by NAICS Industry;Table CA25N Total Full-Time and Part-Time Employment by NAICS Industry.

Table A-8. Production Unit Costs (Facility Specific and Average) per Smolt for Agency Release Strategy at Mitchell Act Hatchery Facilities, by Operating Entity

Table A-8. Smolt Production Unit Costs (Facility Specific and Average) Per Smolt for Agency Release Strategy at Mitchell Act Hatchery Facilities, by Operating Entity								
Agency	**Species**	**Hatchery**	**Release Strategy /2**	**Pounds**	**Operations Costs /1**	**Headquarters /3**	**Capital Costs /4**	**External Costs /5**
ODFW	Coho	Big Creek	535,000	44,583	0.635	0.178	0.040	
ODFW	Coho	Bonneville	4,810,000	289,274	0.336	0.094	0.040	0.332
ODFW	Coho	Sandy	1,000,000	66,667	0.514	0.144	0.040	0.332
	COHO (average)				0.495	0.138	0.040	
ODFW	Fall Ch.	Big Creek	5,700,000	71,250	0.095	0.027	0.040	
ODFW	Spr. Ch.	Clackamas	361,120	36,830	0.651	0.182	0.040	
ODFW	Sum. St.	Bonneville	215,000	43,000	1.592	0.446	0.040	
ODFW	Wint. St.	Big Creek	140,000	17,619	0.969	0.271	0.040	
ODFW	Wint. St.	Bonneville	260,000	43,333	1.328	0.372	0.040	
	WINTER STEELHEAD (average)				1.148	0.321	0.040	
USFWS	Coho	Eagle Cree	2,250,000	90,722	0.348	0.093	0.040	0.571
USFWS	Coho	LWS/Willa	650,000	32,500	0.896	0.239	0.040	
	COHO (average)				0.622	0.166	0.040	
USFWS	Fall Ch.	LWS/Willa	8,200,000	27,079	0.061	0.016	0.040	
USFWS	Fall Ch.	Spring Cre	6,493,000	80,151	0.174	0.047	0.040	
	FALL CHINOOK (average)				0.118	0.031	0.040	
USFWS	Spr. Ch.	Carson	1,420,000	88,750	0.698	0.187	0.040	
USFWS	Spr. Ch.	LWS/Willa	1,000,000	66,667	0.682	0.182	0.040	
	SPRING CHINOOK (average)				0.690	0.184	0.040	
USFWS	Wint. St.	Eagle Cree	100,000	20,000	1.841	0.492	0.040	
WDFW	Coho	Elochoman	915,000	53,824	0.336	0.090	0.040	
WDFW	Coho	Fallert Cre	350,000	20,588	0.309	0.083	0.040	
WDFW	Coho	Kalama Fa	350,000	20,588	0.309	0.083	0.040	
WDFW	Coho	North Fork	800,000	53,333	0.430	0.115	0.040	
WDFW	Coho	Washouga	3,150,000	163,235	0.232	0.062	0.040	
	COHO (average)				0.323	0.086	0.040	
WDFW	Fall Ch.	Elochoman	2,000,000	28,571	0.110	0.029	0.040	
WDFW	Fall Ch.	Fallert Cre	2,500,000	35,714	0.104	0.028	0.040	
WDFW	Fall Ch.	Kalama Fa	2,500,000	35,714	0.104	0.028	0.040	
WDFW	Fall Ch.	North Fork	2,500,000	35,714	0.122	0.033	0.040	
WDFW	Fall Ch.	Ringold Sp	3,450,000	57,500	0.092	0.025	0.040	
WDFW	Fall Ch.	Washouga	4,000,000	61,538	0.089	0.024	0.040	
	FALL CHINOOK (average)				0.104	0.028	0.040	
WDFW	Spr. Ch.	Fallert Cre	125,000	12,500	0.499	0.133	0.040	
WDFW	Spr. Ch.	Kalama Fa	375,000	37,500	0.499	0.133	0.040	
	SPRING CHINOOK (average)				0.499	0.133	0.040	
WDFW	Sum. St.	Elochoman	30,000	6,000	1.044	0.279	0.040	
WDFW	Sum. St.	Fallert Cre	30,000	5,455	2.354	0.629	0.040	
WDFW	Sum. St.	Kalama Fa	60,000	10,909	2.354	0.629	0.040	
WDFW	Sum. St.	North Fork	25,000	4,545	1.099	0.293	0.040	
WDFW	Sum. St.	Ringold Sp	180,000	36,000	0.680	0.182	0.040	
WDFW	Sum. St.	Skamania	254,000	50,800	1.170	0.312	0.040	
	SUMMER STEELHEAD (average)				1.450	0.387	0.040	
WDFW	Wint. St.	Elochoman	105,000	20,818	1.035	0.277	0.040	
WDFW	Wint. St.	Kalama Fa	100,000	16,370	2.123	0.567	0.040	
WDFW	Wint. St.	Skamania	190,000	38,000	1.170	0.312	0.040	
	WINTER STEELHEAD (average)				1.443	0.385	0.040	
Yakama	Coho	Klickitat	1,000,000	66,667	0.134	0.016	0.040	
Yakama	Coho	Yakama N	500,000	25,000	0.209	0.025	0.040	
	COHO (average)				0.171	0.020	0.040	
Yakama	Fall Ch.	Klickitat	4,000,000	53,333	0.044	0.005	0.040	
Yakama	Fall Ch.	Prosser	1,700,000	28,333	0.036	0.004	0.040	
	FALL CHINOOK (average)				0.040	0.005	0.040	
Yakama	Spr. Ch.	Klickitat	600,000	40,000	0.261	0.031	0.040	
TOTAL SMOLT PRODUCTION			64,923,120	2,036,977				

Table A-8 (con't). Smolt Production Unit Costs (Facility Specific and Average) per Smolt for Agency Release Strategy at Mitchell Act Hatchery Facilities, by Operating Entity

	Operations Costs /1	Headquarters /3	Capital Costs /4	External Costs/5
COMPUTED VALUES FOR OTHER OPERATING ENTITIES				
IDFG, CTUIR (Spring Chinook)				
ODFW	0.651	0.182	0.040	
USFWS	0.690	0.184	0.040	
WDFW	0.499	0.133	0.040	
YAKAMA	0.261	0.031	0.040	
AVERAGE	0.525	0.133	0.040	
IDFG (Summer Steelhead)				
ODFW	1.592	0.446	0.040	
WDFW	1.450	0.387	0.040	
AVERAGE	1.521	0.416	0.040	
ALL CHUM (from WDFW cost sheets for Washougal Hat. p	0.055	0.002	0.040	
(Notes: Operations costs include rearing, marking, and administrative costs; capital costs estimated based on ratio [1.03] of capital costs to operating costs for coho at Washougal)				
ALL SOCKEYE (from ODFW cost sheets for Redfish Lake I	1.425	0.008	0.040	
(Notes: Operations costs include program costs minus headquarter costs; capital costs estimated based on ratio [0.65] of capital costs to operating costs for average ODFW coho costs in row 10)				

Notes:
1. Operation costs include fish production, maintenance, marking, onsite monitoring and evaluation, and other costs for hatchery operations in a particular budget year. Fish production costs include labor, feed, materials and services, fish health, fish feed quality control program, annual maintenance, transportation, and administration costs not covered by agency indirect. The particular year may have different production than used in the evaluation of MA EIS alternatives, but the costs per smolt are assumed to apply to the MA EIS alternatives' production.
2. Smolt releases and operation costs per pound are from data generated for Erik White, NOAA Fisheries in May 2008; and by the two states, USFWS, and the Yakama Nation. Chum, cutthroat, and sockeye costs for the MA hatcheries are not shown because they are not used to compare MA EIS alternatives.
3. Hatchery production indirect cost charged by states' central management (i.e. Olympia and Salem headquarters' costs) is 26.7% for WDFW, 28.0% for ODFW, 26.7% for USFWS, and 11.8% for the Yakama Nation.
4. Capital costs are estimated based on research by Carter on the average annualized capital cost for facility replacement at Oregon state-funded hatcheries.
5. Smolts described as external are those reared or partially reared at one or more MA funded hatcheries and transported and released at a non-MA hatchery. There may be other situations where smolts are transported to other hatcheries for final grow-out or to other remote sites for liberation, but costs are included in MA funded operations. When transfers are eggs or parts, external costs include rearing. The per smolt release weight when fish are exported to a non-MA hatchery is assumed to be similar to the on-station weight of releases at the origin hatchery. External acclimation costs use estimates from the SAFE Program for Youngs Bay and Deep River as applicable to other external situations.

Source of Facility Cost Data and Notes: TRG 2009